

AWS Cost Estimator

Pricing & Infrastructure Report

PNB

Mode: SaaS — BusinessNext
Hosted

Date: 24 March 2026

Inflation: 4% / yr

AWS Region: us-east-1

Executive Summary

High-level cost overview and infrastructure snapshot

\$10,158	\$121,901	\$660,255	15,500
Monthly Cost	Annual Cost	5-Year Total	Named Users
Production + environments	Year 1 total	At 4% inflation/yr	Year 1 licensed

Monthly	Annual	5-Year
\$10,158	\$121,901	\$660,255

Project Details

Customer / Bank	PNB
Client Mode	SaaS — BusinessNext Hosted
Database Type	PostgreSQL
Total Named Users	15,500
Concurrent Users	4,650 (30% of named)
Mobile Users	4,050
Worker Nodes	9
Total vCPUs (worker)	96
Total RAM GB (worker)	172 GB
Data Size	2100 GB
Estimate Date	24 March 2026

Node Distribution

AI-assisted infrastructure architecture

Total Worker Nodes	9
Total DB Nodes	11
Fixed / Managed Roles	7
Confidence	High
Distribution Method	Groq LLM + Rule-based

Category	Role / Service	Nodes	Instance Fam.	vCPU/node	RAM GB/node	Model
Kubernetes Cluster	Worker Nodes Linux/RHEL (Web, Mobile, WebAPI)	6	Compute Intensive	16	32	Reserved 3 Yr
Kubernetes Cluster	Worker Nodes Linux/RHEL for Graphana & Prometheus	1	Compute Intensive	8	16	Reserved 3 Yr
Kubernetes Cluster	Worker Nodes Linux/RHEL for EFK - optional	2	Compute Intensive	8	16	Reserved 3 Yr
PGSQL Database	Linux Nodes – Database server (Primary)	2	Memory Intensive	32	128	None
PGSQL Database	Linux Nodes – Cluster (etcd+haproxy, pgbackrest)	4	Memory Intensive	2	8	None
PGSQL Database	Storage: SAN (Primary DB)	2	10K IOPS	—	—	None
S3	S3 Storage + Replication	1	Cloud Storage Service	—	—	On Demand
Security & Compliance	WAF + AWS Shield Advanced	1	Managed Service	—	—	On Demand
Security & Compliance	CloudTrail + Config Audit Logs	1	Managed Service	—	—	On Demand
Kubernetes Cluster	Cloud Managed K8s	1	K8s Service	—	—	None
Caching & Session	Elasticaache Service	2	Memory Optimized	4	16	Reserved 3 Yr
Infrastructure & Monitoring	Load Balancer (Internal + External)	2	ALB	—	—	On Demand
Infrastructure & Monitoring	Bastion Host	1	General Purpose	2	4	Reserved 3 Yr
Infrastructure & Monitoring	NAT Gateway	2	NAT	—	—	On Demand
Infrastructure & Monitoring	Image Registry (ECR)	1	Managed Service	—	—	On Demand
Infrastructure & Monitoring	Back Up - DB & Infra Logs	1	Cloud Storage Service	—	—	On Demand

AWS Cost Breakdown

Service-by-service monthly and annual costs

Compute & Database Roles

Category	Role	Nodes	Instance	\$/hr	\$/month	\$/year
Kubernetes Cluster	Worker Nodes Linux/RHEL (Web, Mobile, WebAPI)	6	m5.4xlarge	\$0.77	\$3,487	\$0
Kubernetes Cluster	Worker Nodes Linux/RHEL for Grafana & Prometheus	1	m5.2xlarge	\$0.38	\$321	\$0
Kubernetes Cluster	Worker Nodes Linux/RHEL for EFK - optional	2	m5.2xlarge	\$0.38	\$724	\$0
PGSQL Database	Linux Nodes – Database server (Primary)	2	r5.8xlarge	\$2.02	\$2,991	\$0
PGSQL Database	Linux Nodes – Cluster (etcd+haproxy, pgbackrest)	4	r5.large	\$0.13	\$400	\$0
PGSQL Database	Storage: SAN (Primary DB)	2	EBS io2	\$0.00	\$525	\$0
S3	S3 Storage + Replication	1	EBS io2	\$0.00	\$738	\$0
Kubernetes Cluster	Cloud Managed K8s	1	EKS	\$0.10	\$73	\$0
Caching & Session	ElastiCache Service	2	cache.r6g.large	\$0.33	\$482	\$0
Infrastructure & Monitoring	Load Balancer (Internal + External)	2	ALB	\$0.00	\$33	\$0
Infrastructure & Monitoring	Bastion Host	1	m5.large	\$0.10	\$94	\$0
Infrastructure & Monitoring	NAT Gateway	2	NAT	\$0.00	\$41	\$0
Infrastructure & Monitoring	Image Registry (ECR)	1	ECR	\$0.00	\$51	\$0
Infrastructure & Monitoring	Back Up - DB & Infra Logs	1	S3	\$0.00	\$118	\$0
Operations	CloudWatch Monitoring	0	CloudWatch	\$0.00	\$80	\$0

Cost Summary

Total Monthly (Production)	\$10,158
Total Annual (Year 1)	\$121,901
5-Year Total (with inflation)	\$660,255
Inflation Rate Applied	4%

5-Year Inflation Forecast

Cost projection at 4% annual inflation

Year	Multiplier	Monthly Cost	Annual Cost	Cumulative Total
Year 1	1.0000×	\$10,158	\$121,901	\$121,901
Year 2	1.0400×	\$10,565	\$126,777	\$248,678
Year 3	1.0816×	\$10,987	\$131,848	\$380,526
Year 4	1.1249×	\$11,427	\$137,122	\$517,648
Year 5	1.1699×	\$11,884	\$142,607	\$660,255

Annual Cost Trend

1		\$121,901
2		\$126,777
3		\$131,848
4		\$137,122
5		\$142,607

Total 5-Year Investment	\$660,255
Average Annual Cost	\$132,051
Average Monthly Cost	\$11,004
Year-over-Year Growth	4.0% (inflation) applied from Year 2

Environment Pricing

Pre-Prod / SIT / UAT and DR environment costs

Environments: Pre-Prod, SIT, UAT

Environments Selected	3 — Pre-Prod, SIT, UAT
Base Cost per Env / mo	\$3,382
Base Cost per Env / yr	\$40,590
Calculation	3 envs × \$3,382/mo = \$10,147/mo total

Monthly	Annual	5-Year
\$10,147	\$121,770	\$0

Category	Role	Nodes	Instance	\$/month
Kubernetes	Cloud Managed K8s	1	EKS	\$73
Kubernetes	Worker Nodes (Web, Mobile, WebAPI)	2	m5.4xlarge	\$1,169
Kubernetes	Worker Nodes (Graphana & Prometheus) – optional	1	m5.2xlarge	\$321
Kubernetes	Worker Nodes (EFK) – optional	1	m5.4xlarge	\$602
PGSQL DB	DB Server (Pre-Prod)	1	m5.4xlarge	\$585
PGSQL DB	Storage: SAN Pre-Prod	1	EBS io2	\$125
PGSQL DB	Storage: SAN SIT	1	EBS io2	\$62
PGSQL DB	Storage: SAN UAT	1	EBS io2	\$62
S3	S3 Storage + Replication	1	S3	\$34
Caching	Caching Service (ElastiCache)	1	cache.r6g.large	\$121
Infrastructure	Load Balancer (Internal + External)	1	ALB	\$16
Infrastructure	NAT Gateway	2	NAT	\$32
Infrastructure	Bastion Host	1	m5.large	\$94
Infrastructure	Image Registry (ECR)	1	ECR	\$50
Backup	Back Up – DB & Infra Logs	1	S3	\$34

Disaster Recovery (DR) Environment

Monthly	Annual	5-Year
\$9,392	\$112,703	\$634,855

Category	Role	Nodes	Instance	\$/month
Kubernetes	Cloud Managed K8s (DR)	1	EKS	\$73
Kubernetes	Worker Nodes (Web, Mobile, WebAPI) – DR	4	m5.4xlarge	\$2,324

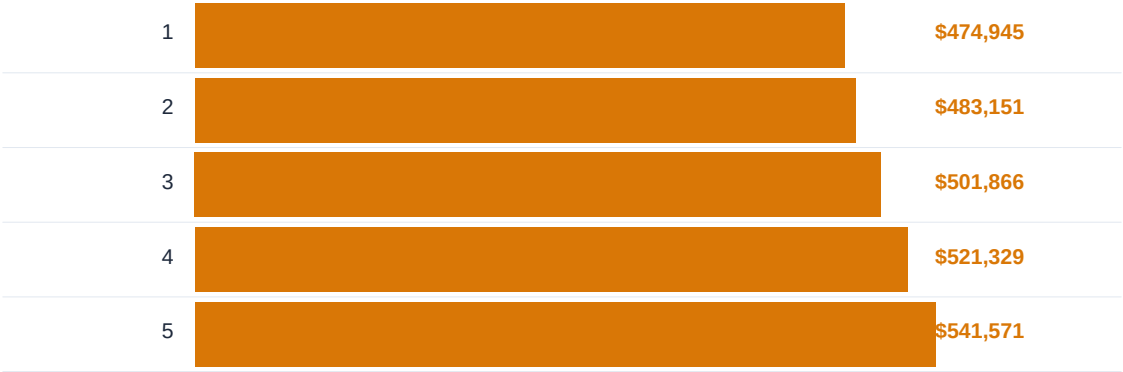
Category	Role	Nodes	Instance	\$/month
Kubernetes	Worker Nodes (Graphana) – DR	2	m5.2xlarge	\$643
Kubernetes	Worker Nodes (EFK) – DR	2	m5.2xlarge	\$724
PGSQL DB	DB Server Primary – DR	2	m5.8xlarge	\$2,291
PGSQL DB	DB Cluster (etcd+haproxy) – DR	4	m5.large	\$312
PGSQL DB	Storage: SAN Primary – DR	2	EBS io2	\$750
PGSQL DB	Reporting DB – DR	1	m5.8xlarge	\$1,145
PGSQL DB	Storage: SAN Reporting – DR	1	EBS io2	\$375
S3	S3 Storage + Replication – DR	1	S3	\$184
Caching	Elasticache Service – DR	2	cache.r6g.large	\$242
Infrastructure	Load Balancer – DR	2	ALB	\$33
Infrastructure	NAT Gateway – DR	2	NAT	\$32
Infrastructure	Bastion Host – DR	1	m5.large	\$94
Infrastructure	Image Registry (ECR) – DR	1	ECR	\$51
Backup	Back Up – DR	1	S3	\$118

PUPM Analysis

Price Per User Per Month — 5-year projection

Year	Named Users	Monthly Usage	Platform Annual	AWS Total	Discounted Cost	PUPM
Year 1	15,500	\$30,706	\$382,891	\$402,036	\$474,945	\$2.5535
Year 2	16,275	\$31,894	\$397,704	\$417,589	\$483,151	\$2.4739
Year 3	17,088	\$33,129	\$413,109	\$433,764	\$501,866	\$2.4475
Year 4	17,943	\$34,414	\$429,130	\$450,587	\$521,329	\$2.4212
Year 5	18,840	\$35,751	\$445,793	\$468,082	\$541,571	\$2.3955

Discounted Annual Cost Trend



Formula Applied

Business Support	3.91% of Total Monthly Usage
Buffer	5% of Total Platform Annual Cost
Managed Services	30% of Total AWS Annual Cost
Discount	11% off Total Cost
PUPM	Discounted Annual Cost ÷ 12 ÷ Named Users
User Growth	+5% per year (Year-over-Year)
One-Time Costs (Y1)	Performance: \$5,000 · Migration: \$5,000 · Managed Setup: \$1,000

GCP Compute Engine Pricing

Google Cloud equivalent — US Central (Iowa)

Monthly	Annual	5-Year
\$15,801	\$189,615	\$1,027,017

Category	Role	Nodes	GCP Machine	\$/hr	\$/month	\$/year
Kubernetes Cluster	Worker Nodes Linux/RHEL (Web, Mobile, WebAPI)	6	n2-standard-16	\$1.17	\$5,378	\$64,540
Kubernetes Cluster	Worker Nodes Linux/RHEL for Grafana & Prometheus	1	n2-standard-8	\$0.58	\$513	\$6,162
Kubernetes Cluster	Worker Nodes Linux/RHEL for EFK - optional	2	n2-standard-8	\$0.58	\$1,201	\$14,412
PGSQL Database	Linux Nodes – Database server (Primary)	2	n2-highmem-32	\$3.86	\$5,734	\$68,804
PGSQL Database	Linux Nodes – Cluster (etcd+haproxy, pgbackrest)	4	n2-highmem-2	\$0.24	\$772	\$9,263
PGSQL Database	Storage: SAN (Primary DB)	2	Persistent Disk Extreme	\$0.00	\$525	\$6,300
S3	S3 Storage + Replication	1	Persistent Disk Extreme	\$0.00	\$738	\$8,850
Kubernetes Cluster	Cloud Managed K8s	1	GKE Standard Cluster	\$0.10	\$73	\$876
Caching & Session	Elasticache Service	2	Memorystore Redis (redis-16gb)	\$0.24	\$358	\$4,292
Infrastructure & Monitoring	Load Balancer (Internal + External)	2	Cloud Load Balancing	\$0.00	\$35	\$420
Infrastructure & Monitoring	Bastion Host	1	n2-standard-2	\$0.15	\$107	\$1,279
Infrastructure & Monitoring	NAT Gateway	2	Cloud NAT	\$0.00	\$32	\$384
Infrastructure & Monitoring	Image Registry (ECR)	1	Artifact Registry	\$0.00	\$51	\$614
Infrastructure & Monitoring	Back Up - DB & Infra Logs	1	Cloud Storage (Nearline)	\$0.00	\$205	\$2,458
Operations	Cloud Operations (Logging + Monitoring)	0	Cloud Operations Suite	\$0.00	\$80	\$960

GCP 5-Year Forecast

Year	Multiplier	Monthly Cost	Annual Cost	Cumulative
Year 1	1.0000×	\$15,801	\$189,615	\$189,615
Year 2	1.0400×	\$16,433	\$197,200	\$386,815
Year 3	1.0816×	\$17,091	\$205,088	\$591,903
Year 4	1.1249×	\$17,774	\$213,291	\$805,194
Year 5	1.1699×	\$18,485	\$221,823	\$1,027,017

AWS vs GCP Cost Comparison

Side-by-side cost analysis — same workload, two clouds

\$10,158	\$15,801	\$5,643	AWS
AWS Monthly	GCP Monthly	Monthly Diff	5-Yr Winner
us-east-1	us-central1	AWS is cheaper	Saves \$366,762

AWS Annual	\$121,901
GCP Annual	\$189,615
AWS 5-Year Total	\$660,255
GCP 5-Year Total	\$1,027,017
Cheaper (Monthly)	AWS
Cheaper (5-Year)	AWS
5-Year Saving	\$366,762

Category-Level Breakdown

Category	AWS \$/mo	GCP \$/mo	Difference	% Diff	✓ Cheaper
Caching & Session	\$482	\$358	\$124	34.7%	GCP
Infrastructure & Monitoring	\$337	\$430	\$92	21.5%	AWS
Kubernetes Cluster	\$4,605	\$7,166	\$2,560	35.7%	AWS
Operations	\$80	\$80	\$0	0.0%	AWS
PGSQL Database	\$3,916	\$7,031	\$3,114	44.3%	AWS
S3	\$738	\$738	\$0	0.0%	AWS
Security & Compliance	\$0	\$0	\$0	0.0%	AWS

5-Year Year-by-Year Comparison

Year	AWS/mo	GCP/mo	AWS/yr	GCP/yr	AWS Cumul.	GCP Cumul.	Winner
Year 1	\$10,158	\$15,801	\$121,901	\$189,615	\$121,901	\$189,615	AWS
Year 2	\$10,565	\$16,433	\$126,777	\$197,200	\$248,678	\$386,815	AWS
Year 3	\$10,987	\$17,091	\$131,848	\$205,088	\$380,526	\$591,903	AWS
Year 4	\$11,427	\$17,774	\$137,122	\$213,291	\$517,648	\$805,194	AWS
Year 5	\$11,884	\$18,485	\$142,607	\$221,823	\$660,255	\$1,027,017	AWS

Notes & Assumptions

Detailed pricing basis, scope, and validity

■ Pricing Validity — 15 Days Only

This estimate is valid for **15 calendar days** from the date of issue (**24 March 2026**). AWS On-Demand prices, instance availability, and currency exchange rates are subject to change without notice. A revised estimate must be requested if requirements change or the validity period lapses.

Pricing & Infrastructure Assumptions

AWS Pricing Basis	All prices are AWS On-Demand, Linux/UNIX, shared tenancy. No Reserved Instances or Savings Plans applied. Switching to 1-Year Reserved pricing typically reduces compute costs by 30-40%.
Region	Estimate uses region: us-east-1. Costs vary by region — a regional multiplier is applied to the base us-east-1 rate.
Inflation Rate	4% per annum applied to all compute and managed-service costs from Year 2 onwards, reflecting typical AWS price adjustments and currency exposure.
High Availability (AZ)	Production and DR environments are architected as Multi-AZ (High Availability). Pre-Prod/SIT/UAT are Single-AZ to optimize costs while maintaining functional parity.
Database Licensing	Postgres is Open-Source (No license cost). SQL Server and Oracle are priced on an AWS RDS 'License Included' basis. Enterprise/BYOL discounts are not applied.
Fallback Rates	If the live AWS Pricing API is unavailable, hardcoded March 2026 on-demand estimates are used. API-sourced prices are marked with a checkmark; estimated prices are noted as estimated.
Node Distribution	Infrastructure sizing computed by a rule-based algorithm, optionally enhanced by Groq LLM (llama-3.3-70b-versatile) at temperature=0. SHA-256 caching prevents redundant API calls.
Named & Concurrent Users	Concurrent users are fixed at 30% of Total Named Users for every forecast year. Year-over-Year growth is applied per the rates selected in the estimator tool.
Security & Compliance Costs	Fixed INR-converted estimates included: Airtel SOC \$500/mo, SOC Machines Rs.30,000/yr, Req 4 Antivirus Rs.2,600/yr, Req 6 Data Discovery Rs.4,000/yr. These are not subject to AWS price changes.
Pre-Prod / SIT / UAT	Environments sized at ~40% of Production infrastructure. PostgreSQL uses self-hosted EC2 (Patroni HA). SQL Server / Oracle use AWS RDS Managed. Costs are additive and not included in the Production PUPM.
Disaster Recovery (DR)	DR sized using a Pilot Light / Warm Standby model at ~50% of Production compute. Active-Active DR would approximately double the DR cost shown.
GCP Pricing	GCP prices are hardcoded us-central1 On-Demand Compute Engine estimates (March 2026). GCP Committed Use Discounts (1-Year CUD) typically reduce costs by ~37%.
PUPM Formula	$PUPM = \text{Discounted Annual Cost} / 12 / \text{Named Users}$ Discounted cost = Total AWS Cost x (1 - 11% discount). Total includes production, DR, Pre-Prod, buffer (5%), Business Support (3.91%), and one-time migration costs in Year 1.
Managed Services	Managed Services fees (30% of AWS annual total) cover platform monitoring, patching, incident management, and lifecycle support by the BusinessNext operations team.

Scope of This Estimate

Included	Production cloud infrastructure (compute, database, storage, networking)
Included	Pre-Production / SIT / UAT environments (if selected)
Included	Disaster Recovery environment (if selected)
Included	BusinessNext platform licensing and managed services
Included	One-time migration, performance testing, and managed-service setup (Year 1)
Excluded	Application development or customisation effort
Excluded	End-user device provisioning or on-site hardware
Excluded	Third-party SaaS subscriptions (e.g. Salesforce, Twilio, Okta)
Excluded	Data centre co-location or physical network charges
Excluded	Regulatory compliance certification fees

Legal Disclaimer

This document is prepared by BusinessNext for the exclusive use of the named client organisation. All cost figures are estimates based on current AWS/GCP published pricing and the inputs provided at the time of generation. Actual costs may vary depending on usage patterns, reserved-instance commitments, enterprise discount programmes, and any changes to cloud provider pricing.

This estimate does **not** constitute a binding commercial offer or contract. A formal Statement of Work (SOW) and Order Form must be executed prior to commencement of any services. BusinessNext reserves the right to revise this estimate upon receipt of updated requirements or upon expiry of the 15-day validity period.

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